

TEAM CONTRIBUTION STATEMENT

Nguyễn Quang Minh-202414642 – System Implementation & Processing

Mainly focused on the **technical implementation** of the system:

- Designed and implemented the full processing pipeline:
Input → Separation → Compression → Reconstruction → Evaluation
- Handled **audio separation** using tools like Audacity and supported model-based approaches (e.g., Demucs if needed)
- Implemented **compression process** using FFmpeg, including testing different bitrate settings
- Developed core Python scripts:
 - reconstruct.py (audio reconstruction using pydub)
 - show_bitrate.py (bitrate calculation using FFmpeg tools)
- Organized project structure, file handling, and ensured code runs without errors

Hà Quang Minh-202414646 – Analysis, Evaluation & Documentation

Mainly focused on **analysis and reporting**:

- Implemented audio analysis using librosa:
 - Waveform visualization
 - RMS energy (compression level)
 - Spectrogram (frequency content)
- Designed and ran **evaluation experiments**:
 - Compared original vs reconstructed audio
 - Compared different audio types (speech vs music)
 - Analyzed bitrate and compression effects
- Interpreted results and wrote key parts of the **project report**:
 - Methodology
 - Evaluation

- Results & Analysis
- Prepared materials for **live demo**, including explanation and presentation flow

Shared Responsibilities

Both members worked together on:

- Testing the full pipeline end-to-end
- Debugging errors and improving code reliability
- Selecting suitable audio samples (dataset)
- Running experiments and verifying results
- Preparing the final demo and presentation

Equal Contribution Confirmation

Although tasks were divided into different roles, both members contributed significantly to the project. Each person was responsible for important parts and also supported the other when needed.

The team worked collaboratively throughout the project, and the final result reflects a **balanced and equal contribution** from both members.